

# Appendix 1

## Practical Guide to KITL/FLT3L Culture (Related to Chapter 1)

### Choosing an *in vitro* culture system for murine dendritic cells (DCs)

Please refer to the practical recommendations in the table below (Table 1)

**Table 1**

| Experimental goal                    | Recommended system                                          |
|--------------------------------------|-------------------------------------------------------------|
| Large scale cDC1 production          | MutuDC cell line (Fuentes Marraco et al., 2012)             |
| Primary cDC1 generation              | KITL/FLT3L (Ou et al., 2023) or iCD103 (Mayer et al., 2014) |
| Multiple DC subsets                  | KITL/FLT3L                                                  |
| Simplest primary culture (low yield) | FLT3L (Brasel et al., 2000)                                 |

### Plating density and medium volume

I recommend plating 6.25 k ACK-lysed BM cells per mL medium. Practically, this corresponds to:

- 500k cells in 0.8 mL medium in a 24-well
- 2.5M cells in 4 mL medium in a 6-well
- Adjust accordingly for other formats

### Recombinant KITL (SCF) and FLT3L

- KITL (SCF): Recombinant KITL (SCF) works (see Supporting Information Figure 1C). I recommend using 200 ng/mL Mouse SCF Recombinant Protein (PeproTech)
- FLT3L: Although I have not tested recombinant FLT3L in the KITL/FLT3L system, 100 ng/mL recombinant FLT3L (PeproTech) is widely used in FLT3L-alone cultures and is very likely effective

### Sources of conditioned media

- Human Flt3L-Fc conditioned media: J558L Cell line from Dr. Marina Cella (mcella@wustl.edu)
- Murine KITL conditioned media: CHO Cell line from Dr. Kyunghee Choi (kchoi@wustl.edu)

If requesting these cell lines, please explain your intended use and include Ken (kmurphy@wustl.edu) in the correspondence.

However, I recommend starting with recombinant proteins for pilot experiments, as they are more readily accessible.

### Medium changes

Two approaches work equally well in my hands:

- In-plate spin: Briefly centrifuge the culture plate, carefully aspirate the old medium without tilting the plate, and add fresh medium
- Tube transfer: Transfer culture contents into a conical tube, centrifuge, discard old medium, resuspend in fresh medium, and return cells to the original well

Note: There is no need to expand culture volume throughout the culture period.

### **Duration of the culture for sorting cDC1s**

I recommend a total of 9-10 days of culture to ensure that all cDC1s have fully upregulated XCR1

### **Expected yield of cDC1s per mouse**

- On average, the KITL/FLT3L protocol generates ~0.5 cDC1s per input ACK-lysed BM cell (see Figure 1C)
- From one mouse, I typically obtain 50–100 M BM cells, which yields ~25-50 M cDC1s
- Yield varies depending on conditions (e.g. recombinant proteins vs. conditioned media), but a conservative expectation is  $\geq 10$  M cDC1s per mouse. *Note that this is still a huge number of cells, usually more than sufficient for most downstream experiments*

### **Removing contaminating macrophages**

Contaminating macrophages can be eliminated by gating out Ly-6C<sup>+</sup> cells during cell sorting.

This is only necessary when sorting cDC2s and not required when sorting cDC1s

### **References**

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